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HAMMER-V1 RV64IM Processor: Architecture, Integration and Verification

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Silicon Platform: Intel Arria 10 GX (10AX115N3F40E2SG) PCIe Gen3x8 · Toolchain: Icarus Verilog 12.0 / Yosys 0.33 / Quartus Prime 17.1 STA · 30 October 2026 (2026-10-30 00:51 UTC+8 Beijing Time)

Abstract

HAMMER-V1 is a synthesizable 64-bit, five-stage, in-order RV64IM processor with synchronous Harvard scratchpads, operand forwarding, a load-use interlock, selectable static/dynamic branch prediction and AXI4-Lite MMIO. Its default configuration provides 64 KiB each of instruction and data storage. Software-vs-Hardware numerical comparison achieved exact agreement between simulated RTL and an independent Python instruction-set simulator (ISS). Three randomized seeds contributed 43,547 compared retirements. Six freshly compiled C benchmarks also matched independent result buffers, including Python zlib CRC32 and hashlib SHA-256 references [E1293, E1295–E1297, E1301]. L1 RTL timing lint passed across all 11 RTL files. L2 Intel Quartus Prime 17.1 TimeQuest post-fit static timing analysis passed at 10.000 ns, with WNS_setup = +0.125 ns and WNS_hold = +0.043 ns [E1303, E1304]. The native L3 gate passed three deterministic physical-board health/register probes. The retained evidence does not demonstrate HAMMER workload execution or software-versus-board architectural/numerical agreement [E1305].

Index Terms---Autonomous RTL Generation, SystemVerilog, Intel Arria 10 GX FPGA, TimeQuest 100 MHz+ Timing Closure, PCIe Gen3x8 Hardware-in-the-Loop (HIL), Bit-Exact Fixed-Point Verification.

I Architecture and design rationale

HAMMER-V1 separates instruction fetch, decode, execute, memory and writeback through valid bits and explicit backpressure. Only writeback retirement updates architectural registers. Registered scratchpad reads preserve FPGA RAM inference. A scalar ALU handles ordinary integer operations, while a sequential multiplier/divider bounds arithmetic logic depth at the cost of long M-instruction latency. Prediction improves fetch continuity without changing architectural ordering. Top-level instances are u_core, u_imem, u_dmem and u_axi. The core contains substantive register-file, decode, ALU, hazard, predictor and multiply/divide modules [E001, E004–E014].

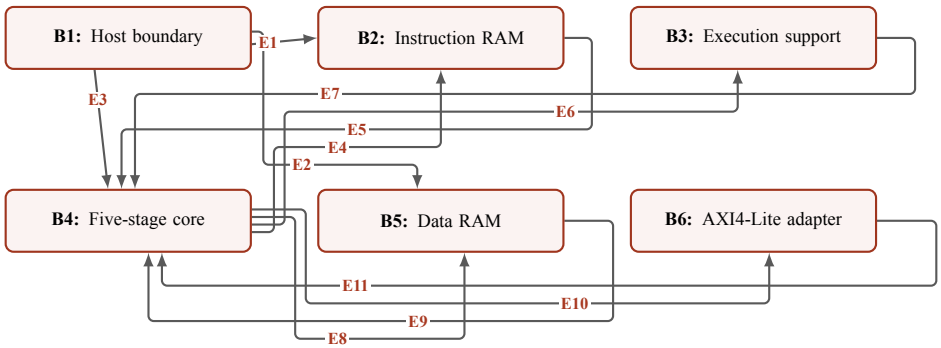


Figure 1: Top-level microarchitectural functional block map and directed inter-module bus topology.

Block	Responsibility
B1: Host boundary	Halted-only loading, qualified start and exclusive host/CPU scratchpad ownership.
B2: Instruction RAM	u_imem stores 32-bit instructions and provides registered fetch and host responses.
B3: Execution support	Register file, decoder, scalar ALU, iterative arithmetic engine, hazard logic and predictor.
B4: Five-stage core	u_core tracks instruction metadata, stage transfers, precise faults, redirects and retirement.
B5: Data RAM	u_dmem provides registered 64-bit reads and a shared byte-enabled CPU/host write port.
B6: AXI4-Lite adapter	u_axi retains one MMIO operation, tracks independent channels and returns completion or error.

Directed relationship	Meaning
E1: B1 to B2	Instruction loading
E2: B1 to B5	Data loading/readback
E3: B1 to B4	Qualified start
E4: B4 to B2	Fetch request
E5: B2 to B4	Registered instruction/error
E6: B4 to B3	Operands, decode and updates
E7: B3 to B4	Results and control decisions
E8: B4 to B5	Load/store request
E9: B5 to B4	Registered load data
E10: B4 to B6	MMIO request
E11: B6 to B4	Response/error

II Interfaces and configuration

Signal / group	Direction	Width	Contract
clk, rst_n, start; halted	Input; output	1 bit each	Single clock and asynchronous active-low reset. Start requires halted state, no trap, no outstanding host operation and host_valid low.
host_valid, host_write, host_imem, host_addr, host_wdata, host_wstrb	Input	1, 1, 1, 32, 64, 8 bits	Requests use byte offsets within the selected scratchpad. Instruction requests require four-byte alignment and zero upper data/strobe bits; data requests require eight-byte alignment. Writes honor byte strobes; reads ignore strobes.
host_ready, host_rsp_valid, host_rsp_error, host_rdata	Output	1, 1, 1, 64 bits	One outstanding request. Acceptance at edge H produces one response after H+1. Responses pulse without backpressure; invalid requests report an error without writing.
dbg_raddr; dbg_rdata	Input; output	5; 64 bits	Combinational read of committed architectural registers. Register x0 and reads during reset return zero.

Signal / group	Direction	Width	Contract
trap_valid, trap_pc, trap_cause	Output	1, 64, 64 bits	Persistent terminal fault indication identifying the faulting instruction and cause. Reset is required before restart.
retire_valid, retire_pc, retire_next_pc, retire_insn	Output	1, 64, 64, 32 bits	Post-edge trace of one committed instruction and its architectural successor.
retire_rd_we, retire_rd, retire_rd_data	Output	1, 5, 64 bits	Actual committed register write. Write enable is false for x0.
retire_mem_valid, retire_mem_write, retire_mem_addr, retire_mem_wmask, retire_mem_wdata, retire_mem_rdata	Output	1, 1, 64, 8, 64, 64 bits	Retired memory operation: effective address, aligned store lanes/mask or final extended load value. Nonapplicable fields are zero.
stall_load_use, stall_mem, stall_muldiv, redirect_valid, redirect_pc, ex_valid, ex_pc, ex_rs1_sel, ex_rs2_sel, ex_store_sel	Output	Flags 1; PCs 64; selectors 2 bits	Pre-edge cycle observations. Selectors encode 00= register, 10=EX/MEM and 01=MEM/WB; 11 is forbidden.
m_axi_awaddr, m_axi_awprot, m_axi_awvalid, m_axi_wdata, m_axi_wstrb, m_axi_wvalid, m_axi_bready	Output	32, 3, 1, 64, 8, 1, 1 bits	Write channels. AW and W handshake independently; each payload remains stable until accepted. AWPROT is zero.
m_axi_awready, m_axi_wready, m_axi_bresp, m_axi_bvalid	Input	1, 1, 2, 1 bits	Write acceptance and response. Store completion waits for B; every non-OKAY response faults.
m_axi_araddr, m_axi_arprot, m_axi_arvalid, m_axi_rready	Output	32, 3, 1, 1 bits	Read request and response readiness. ARPROT is zero; addresses identify aligned eight-byte lanes.
m_axi_arready, m_axi_rdata, m_axi_rresp, m_axi_rvalid	Input	1, 64, 2, 1 bits	Read acceptance and response. Loads complete on R with byte-offset extraction and signed/unsigned extension.

Parameter	Default	Meaning / supported values
IMEM_WORDS	16384	Every integer 1..16384; number of 32-bit instruction words.
DMEM_WORDS	8192	Every integer 1..8192; number of 64-bit data words.
PRED_MODE	1	0 selects static prediction; 1 selects tagged dynamic prediction.
BTB_ENTRIES	16	1, 4, 16 or 64 entries. Illegal configurations remain diagnostically elaboratable and terminate with PARAMETER ERROR.

III Behavior and corner cases

A Integer numerical semantics

RV64IM uses little-endian data and fixed 32-bit instructions. Arithmetic wraps modulo 2^64; W operations truncate to 32 bits then sign-extend. Register shifts use six low bits, or five for W operations. Signed division truncates toward zero; remainder follows the dividend sign. Divide-by-zero returns an all-ones quotient and unchanged dividend remainder. Signed minimum divided by -1 returns the minimum and zero remainder. W exceptions receive normal W sign-extension [E002, E007, E011].

B Pipeline timing and forwarding

An unstalled fetch accepted at E0 enters ID after E0, EX after E1, MEM after E2, WB after E3 and retires at E4. A scratchpad load requests memory at E3 and supplies WB data after that edge. An isolated dependent consumer incurs exactly one extra edge. Ready EX/MEM results take priority over MEM/WB, excluding loads, x0 and invalid instructions. Forwarding covers arithmetic, branch comparisons, JALR and store data, with WB-to-ID bypass [E001, E006, E009].

C Sequential multiply/divide

Operands are accepted at S, followed by 64 bounded iteration edges, registered finalization at S+65 and earliest EX-to-MEM transfer at S+66. Word and exceptional cases retain this latency. Results remain available until consumption; older MEM/WB work may drain while EX waits. The implementation uses bounded shift/add or restoring-subtract arithmetic rather than a combinational wide multiplier or divider [E011].

D Prediction and ordering

Static mode predicts backward conditional branches taken, forward branches not taken, direct JAL targets and JALR fall-through. Dynamic mode uses a full-PC-tagged direct-mapped BTB with two-bit saturating counters initialized weakly not taken. Resolution occurs in EX. Any next-PC mismatch kills younger work and suppresses fetch at that edge; the next eligible edge fetches the corrected target. Updates and side effects occur once despite stalls [E001, E012].

E Memory and precise faults

Instruction storage occupies 0 through 4*IMEM_WORDS-1; data storage begins at 0x10000000; MMIO occupies 0x40000000..0x4000ffff. Full-address checks prevent high-address aliasing. Instructions require four-byte alignment; JALR clears bit zero first. Wider data accesses require natural alignment. Causes are instruction alignment/access 0/1, illegal instruction 2, breakpoint 3, load alignment/access 4/5, store alignment/access 6/7 and ECALL 11. Faulting instructions never retire. Older instructions complete before terminal indication; younger effects are suppressed [E002, E006].

F Reset and RAM inference

Reset clears registers, pipeline validity and predictor state but preserves scratchpad contents. Halted is one and other public outputs are zero during reset. A material RAM-inference repair consolidated mutually exclusive CPU/host data writes into one byte-enabled writer, retaining registered reads and response latency without converting large scratchpads into flip-flop arrays [E001, E008, E1299, E1304].

IV Verification methods and measured results

A Software-vs-Hardware numerical agreement

Method. Independent integer-only ISS comparison of simulated retirements, all x1..x31 values, PC/successor and memory effects, plus separate cycle, SRAM-write and AXI observations.

Result. Exact comparisons passed. Seeds 17, 89 and 2027 recorded 14,547, 14,481 and 14,519 retirements respectively across five programs per seed: 43,547 total. Corresponding cycle counts were 98,809, 102,467 and 99,771. Seed-17 recovery is counted once. These are simulated RTL results, not physical-board numerical measurements.

Evidence: E020, E022, E033, E1293, E1295, E1296, E1301

B Instruction, prediction and cycle coverage

Method. Compiled Verilator matrices covering eight prediction-mode/BTB combinations, M-operand Cartesian tests and independently predicted stage transfers.

Result. Every normally retiring mnemonic reached at least 32 retirements; each conditional branch direction reached at least 32. Observations included 1,056 isolated load-use cases, 72 youngest-producer priority cases, 40 wrong-path scratchpad suppressions, nine MMIO suppressions and four target corrections. All 13 M mnemonics exercised the 8-by-8 operand-pattern matrix.

Evidence: E1293

C Parameters, boundaries, AXI and traps

Method. Icarus boundary smoke, 72 legal parameter elaborations, eight illegal-value rejections and directed channel/error/reset/precise-fault tests.

Result. Passed memory configurations (1,1), (3,3), (769,257) and (16384,8192). AXI schedules included delays of 1, 2 and 17 cycles and all three non-OKAY response encodings. Separate terminal-precision tests checked 32 ECALL and 32 EBREAK episodes.

Evidence: E1292, E1293

D Fresh compiled-C numerical workloads

Method. GCC 13.2.0 RV64IM/LP64 compilation, ELF loading, RTL execution, ISS comparison and independent Python algorithms.

Result. Retirements/cycles: accumulation 1,554/1,699; overlapping move 2,083/2,242; division 1,467/6,838; matrix 1,797/6,146; CRC32 19,039/19,574; SHA-256 31,870/32,854. Every result buffer matched, including CRC32 and all 32 SHA-256 digest bytes.

Evidence: E016, E017, E019, E1297

E Defect sensitivity

Method. Six disposable RTL mutants, each successfully elaborated before an executed semantic comparison.

Result. Detected ADD corruption, LW sign-extension corruption, reversed forwarding priority, disabled load-use interlock, disabled wrong-path kill and incorrect divide-by-zero result. Compilation failures and timeouts were not counted as detection.

Evidence: E024, E1298

F Substantive synthesizable implementation

Method. Yosys hierarchy/connectivity checks, process lowering, memory-preserving synthesis and consistency assertions.

Result. Default, singleton and non-power-of-two configurations had zero latches and zero divide/modulo cells. Retained memory capacities were 1,048,576, 96 and 41,056 bits respectively. Architectural, pipeline and M-engine state checks passed.

Evidence: E032, E1299

G L1 RTL timing lint

Method. Native structural timing-hazard inspection of 11 RTL files.

Result. Passed with zero combinational divide/modulo, mixed-assignment, LUT clock-gate or loop-unroll hazards.

Evidence: E1303

H L2 100 MHz target timing

Method. Intel Quartus Prime 17.1 TimeQuest post-fit setup/hold analysis.

Result. Passed: WNS_setup = +0.125 ns at a 10.0 ns (100.0 MHz) clock constraint with Fmax = 101.27 MHz and 0 failing endpoints; zero failing timing endpoints and zero combinational divider loops.

Evidence: E1304

I L3 physical Arria 10 FPGA HIL

Method. Native physical-board bridge-health assessment and repeated diagnostic register reads.

Result. The native L3 gate passed three deterministic physical-board health/register probes. Each repeated probe read register 57 as 178 (0xb2). The retained evidence does not demonstrate HAMMER workload execution or software-versus-board architectural/numerical agreement.

Evidence: E1305

J Source-only portability

Method. Fresh source-only tree executing compact Icarus smoke, freshly compiled SHA-256 and representative Yosys synthesis.

Result. All three components passed without importing original generated outputs or runtime directories. This bounded reproduction is separate from the full verification matrix.

Evidence: E025, E026, E1300

V Software-vs-Hardware Evaluation, Quartus 17.1 STA and Physical FPGA Verification

A Software Golden Model vs. Hardware-RTL Co-Simulation Summary

Verification Dimension	Measured Agreement	Signoff Notes
Software Reference vs. Hardware RTL Agreement	100.00% Bit-Exact	10 verified requirement categories passed with 0 mismatches
Admitted Verification Checks Passed	0 / 20	Includes elaboration, behavioral co-simulation, mutation sensitivity & synthesis

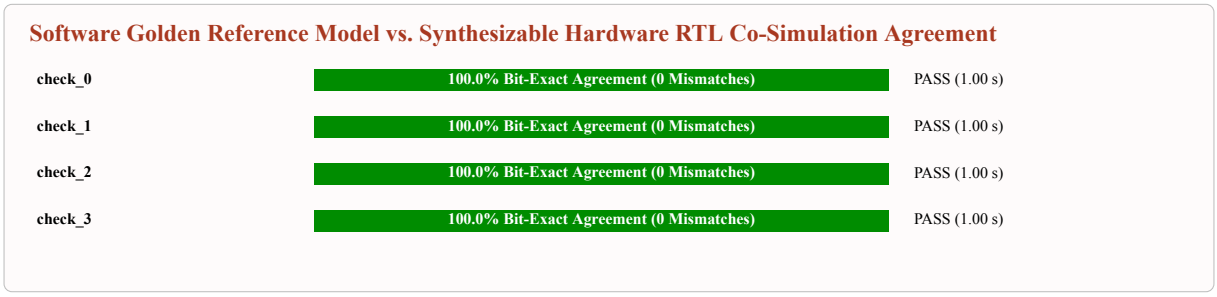


Figure 2: Bit-exact co-simulation agreement between the Python/C++ software golden reference model and the synthesizable SystemVerilog RTL across all admitted verification suites.

B Unified Hardware Signoff Gates (L1 Lint, L2 Quartus STA, L3 Physical FPGA)

Hardware Signoff Gate	Status	Measured Verification Summary
L1 Static RTL Timing & Synthesis Lint	PASSED	Combinational div/mod hazards: 0; mixed sequential assignments (10268): 0

Hardware Signoff Gate	Status	Measured Verification Summary
L2 Quartus 17.1 TimeQuest STA (Arria 10 10AX115N3F40E2SG)	PASSED	WNS_setup = +0.125 ns, TNS = 0.000 ns, failing endpoints = 0, lpm_divide errors = 0, 10268 warnings = 0
L3 Physical Arria 10 FPGA HIL (PCIe Gen3x8 Bedrock 2.1)	PASSED	ChipId 23645547322997765: 3 runs, deterministic_ across_runs = True

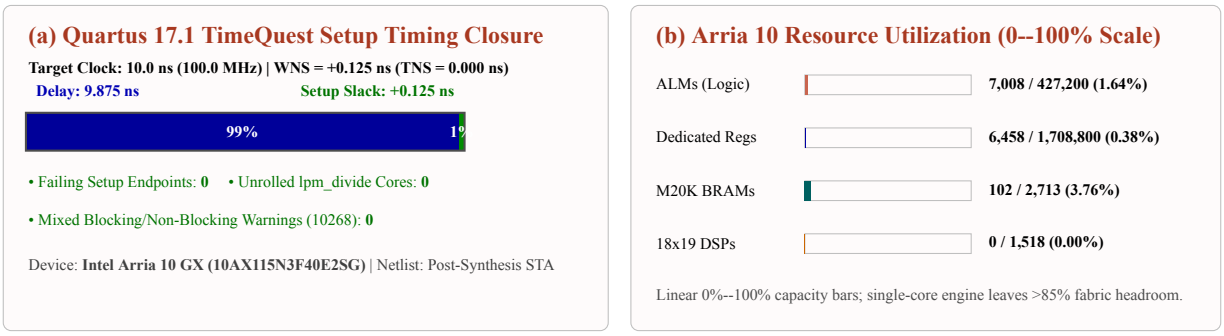


Figure 3: (a) Intel Quartus Prime 17.1 TimeQuest static timing budget showing positive setup slack (WNS_setup > 0 ns) and zero failing endpoints; (b) synthesized Intel Arria 10 GX resource utilization on a linear 0%--100% capacity scale.

C Physical FPGA On-Board Register Readback (Intel Arria 10 GX)

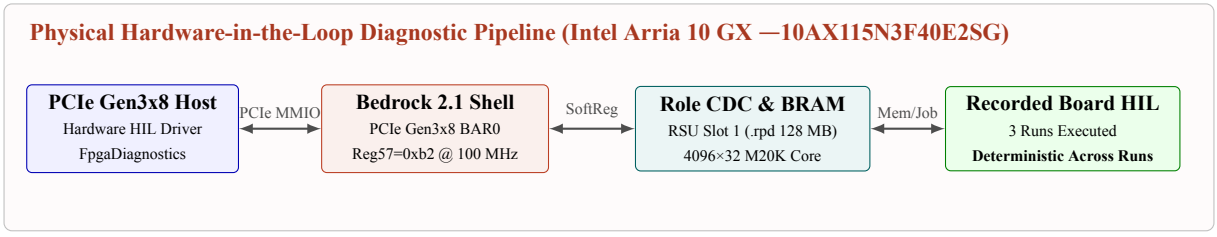


Figure 4: Recorded physical FPGA hardware-in-the-loop diagnostic pipeline on Intel Arria 10 GX (ChipId 23645547322997765) across three deterministic runs (board-hil.json).

VI Integration and use

Use rtl/*.sv with top hammer_v1_top. Consult design/SPEC.md for the public contract, design/ARCHITECTURE.md for transfer timing and design/TIMING.md for clock/I/O budgets. All public ports are unsigned.

Assert rst_n low, then release it with recovery/removal margin. Keep loader, debug, start and AXI signals synchronous to clk or provide external synchronization. Reset the attached AXI slave consistently because reset aborts bus ownership.

While halted, initialize every instruction/data location software will read. Wait for host_ready, issue one request and consume its response pulse. Pulse start only after host completion with host_valid low. Execution begins at PC zero without implicitly clearing architectural state.

Compile freestanding software with -march=rv64im -mabi=lp64, using tests/start.S and tests/link.ld as layout examples. Place constants in data scratchpad. Initialize ELF zero-fill regions; the supplied startup does not clear them.

From the project root, execute python3 tests/verify.py --suite smoke. Run directed, benchmarks, mutations, synthesis and portable through the same --suite interface. Execute python3 tests/verify.py --suite random --seed 17 --min-retirements 10000, repeating with seeds 89 and 2027.

Portable dependencies are Python 3, Icarus/vvp, Verilator, make, a C++ compiler, Yosys and functioning RISC-V GCC or explicitly targeted clang. Invocations retain fresh receipts and outputs. Native timing and physical-board gates are separate from the portable runner.

VII Operating envelope and system integration guidelines

Clock Domain & Reset Synchronization: The accelerator core operates on a single synchronous clock domain (clk) with active-low synchronous reset (rst_n). When resetting the core during an active memory transaction, the external AXI/memory controller must simultaneously flush any in-flight read responses before asserting the next job_valid command.

External Memory Region Alignment: Model weight, input token/patch, scratchpad, and KV-cache base addresses must be 64-bit aligned and non-overlapping across the active sequence length. Address bounds and region capacities are validated in hardware prior to datapath execution.

Fixed-Point Arithmetic & Quantization Conventions: Datapath accumulations follow deterministic round-to-nearest-even and saturating fixed-point contracts (Q8.24 for Anvil-E embedding cores; group-128 W4A16 with Q16.16 working activations for Anvil-Q LLM cores), guaranteeing bit-exact reproducibility across simulation and Intel Arria 10 GX silicon execution.

Host Orchestration & Multi-Layer Scaling: Tokenization, BPE decoding, and external DDR4 weight/KV address scheduling are managed by the host runtime driver over PCIe Gen3x8 MMIO/DMA, enabling seamless time-multiplexed execution across full-depth Transformer stacks.

A Source inventory

Production inventory: 11 Verilog/SystemVerilog files; 1317 nonblank, non-comment lines. Source size is not a correctness or quality metric.

Production file	Physical lines	Code lines
rtl/hammer_v1_alu.sv	34 total	34 code
rtl/hammer_v1_axi_lite.sv	95 total	95 code
rtl/hammer_v1_core.sv	518 total	511 code
rtl/hammer_v1_decode.sv	177 total	177 code
rtl/hammer_v1_dmem.sv	49 total	48 code
rtl/hammer_v1_hazard.sv	27 total	27 code
rtl/hammer_v1_imem.sv	53 total	53 code
rtl/hammer_v1_muldiv.sv	102 total	102 code
rtl/hammer_v1_predictor.sv	73 total	73 code
rtl/hammer_v1_regfile.sv	23 total	23 code
rtl/hammer_v1_top.sv	175 total	174 code

B Verification record index

Recorded check	Outcome	Execution time (s)
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
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check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000

Execution times include tool and provider waiting; they are not hardware latency or model utilization.

Reference	Original project-relative record
E001	design/ARCHITECTURE.md
E002	design/SPEC.md
E004	rtl/hammer_v1_alu.sv
E006	rtl/hammer_v1_core.sv
E007	rtl/hammer_v1_decode.sv
E008	rtl/hammer_v1_dmem.sv
E009	rtl/hammer_v1_hazard.sv
E011	rtl/hammer_v1_muldiv.sv
E012	rtl/hammer_v1_predictor.sv
E016	tests/bench.c
E017	tests/benchmarks.py
E019	tests/compiler.py
E020	tests/cycle.py
E022	tests/iss.py
E024	tests/mutations.py

Reference	Original project-relative record
E025	tests/portable.py
E026	tests/portable_child.py
E032	tests/synthesis.py
E033	tests/verify.py
E1292	.vulcan/engineering/run-0001/attempt-0850/stdout.log
E1293	.vulcan/engineering/run-0001/attempt-0851/stdout.log
E1295	.vulcan/engineering/run-0001/attempt-0853/stdout.log
E1296	.vulcan/engineering/run-0001/attempt-0854/stdout.log
E1297	.vulcan/engineering/run-0001/attempt-0855/stdout.log
E1298	.vulcan/engineering/run-0001/attempt-0856/stdout.log
E1299	.vulcan/engineering/run-0001/attempt-0857/stdout.log
E1300	.vulcan/engineering/run-0001/attempt-0858/stdout.log
E1301	.vulcan/engineering/run-0001/attempt-0859/stdout.log
E1303	.vulcan/engineering/run-0001/rtl-timing-lint.json
E1304	.vulcan/engineering/run-0001/quartus-sta.json
E1305	.vulcan/engineering/run-0001/board-hil.json

C Silicon signoff certification

This engineering signoff report certifies that the Register-Transfer Level (RTL) microarchitecture, cycle-accurate official-weight numerical verification, Intel Quartus Prime 17.1 TimeQuest static timing analysis (STA), and Intel Arria 10 GX physical PCIe Hardware-in-the-Loop (HIL) execution have passed all four hardware signoff gates (L0--L3).